DermaLens eye-bag training (Colab)

Before running:

1. Locally: `python scripts/make_colab_bundle.py --version v1` and upload the zip to your Google Drive root.
2. Runtime -> Change runtime type -> **T4 GPU**.
3. Run cells top to bottom. If the session disconnects mid-training, just re-run — the train cell auto-resumes from `last.pt`.

Expected runtimes on a T4 with ~1.5k crops: binary ~10-20 min, ordinal ~20-35 min.

```
BUNDLE_VERSION = 'seed'   # bump after re-annotation / re-split
# Stages: seed_pretrain (throwaway pre-annotator; bundle with --splits data/seed_splits)
#         -> baseline_binary -> ordinal_severity -> multitask
STAGE = 'seed_pretrain'

from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
# Unpack the bundle and restore any previous experiments from Drive
import os, pathlib, shutil

WORK = '/content/dermalens'
DRIVE_DIR = '/content/drive/MyDrive/dermalens'
bundle = f'/content/drive/MyDrive/dermalens_bundle_{BUNDLE_VERSION}.zip'
assert os.path.exists(bundle), f'Upload {bundle} to your Drive root first'

os.makedirs(WORK, exist_ok=True)
os.makedirs(DRIVE_DIR, exist_ok=True)
!cd {WORK} && unzip -qo {bundle}

# Restore experiments/ from Drive so --resume works across sessions
drive_exp = f'{DRIVE_DIR}/experiments'
if os.path.exists(drive_exp):
    shutil.copytree(drive_exp, f'{WORK}/experiments', dirs_exist_ok=True)
    print('Restored experiments/ from Drive')
%cd {WORK}
```

```
/content/dermalens
```

```
# Dependencies: torch/torchvision/pandas/sklearn are Colab-preinstalled.
# The training import chain needs only these two extras (no mediapipe/cv2).
!pip -q install albumentations pyyaml
import torch; print('CUDA:', torch.cuda.is_available(), torch.cuda.get_device_name(0) if tor
```

```
CUDA: True Tesla T4
```

```
# Sanity check: 2-epoch smoke run on the bundled synthetic data (~1 min)
!python scripts/train.py --config configs/smoke.yaml --smoke
```

```
02:58:42  INFO     SMOKE TEST MODE: small encoder, 2 epochs, no pretrained download
/usr/local/lib/python3.12/dist-packages/albumentations/core/validation.py:114: UserWarning: S
  original_init(self, **validated_kwargs)
02:58:42  INFO     Dataset loaded: 84 crops from data/smoke/splits/train.csv  grade counts: {
02:58:42  INFO     Dataset loaded: 14 crops from data/smoke/splits/val.csv  grade counts: {0:
02:58:42  INFO     Sampler class counts (severity): {0: 20, 1: 16, 2: 12, 3: 24, 4: 12}
02:58:42  INFO       class 0:    20 samples → 22.0% of each epoch (raw share: 23.8%)
02:58:42  INFO       class 1:    16 samples → 19.7% of each epoch (raw share: 19.0%)
02:58:42  INFO       class 2:    12 samples → 17.1% of each epoch (raw share: 14.3%)
02:58:42  INFO       class 3:    24 samples → 24.1% of each epoch (raw share: 28.6%)
02:58:42  INFO       class 4:    12 samples → 17.1% of each epoch (raw share: 14.3%)
02:58:42  INFO     EyeBagModel: encoder=resnet18  params=11.4M  heads=[presence, severity, da
02:58:42  INFO     Trainer ready: device=cuda  amp=False  epochs=2  monitor=val_qwk
02:58:46  INFO     Epoch   1/2  train_loss=1.3046  val_qwk=0.0000  lr=1.20e-05  (3s)  ★ best
02:58:47  INFO     Epoch   2/2  train_loss=1.3522  val_qwk=0.0000  lr=2.20e-05  (1s)
02:58:47  INFO     Training complete. Best val_qwk=0.0000 @ epoch 1
02:58:47  INFO     Checkpoints in experiments/smoke/  (best.pt, last.pt)
02:58:47  INFO     Final validation metrics:
02:58:47  INFO       accuracy       : 0.7143
02:58:47  INFO       auprc          : 0.6755
02:58:47  INFO       auroc          : 0.4500
02:58:47  INFO       brier          : 0.2120
02:58:47  INFO       dc_accuracy    : 0.4286
02:58:47  INFO       dc_auprc       : 0.7469
02:58:47  INFO       dc_auroc       : 0.7083
02:58:47  INFO       dc_brier       : 0.2520
02:58:47  INFO       dc_ece         : 0.0907
02:58:47  INFO       dc_f1          : 0.0000
02:58:47  INFO       dc_precision   : 0.0000
02:58:47  INFO       dc_sensitivity : 0.0000
02:58:47  INFO       dc_specificity : 1.0000
02:58:47  INFO       ece            : 0.0699
02:58:47  INFO       exact          : 0.1429
02:58:47  INFO       f1             : 0.8333
02:58:47  INFO       loss           : 1.4121
02:58:47  INFO       mae            : 1.2857
02:58:47  INFO       precision      : 0.7143
02:58:47  INFO       qwk            : 0.0000
02:58:47  INFO       sensitivity    : 1.0000
02:58:47  INFO       specificity    : 0.0000
02:58:47  INFO       within_one     : 0.5714

Next step:
  python scripts/error_analysis.py --checkpoint experiments/smoke/best.pt --csv data/smoke/sp
```

```
# Train the selected stage, auto-resuming if a previous session left a last.pt.
# Stage order and warm-starts:
#   0. seed_pretrain     (throwaway pre-annotator, no warm-start)
#   1. baseline_binary   (Milestone 1, monitor val_auroc)
#   2. ordinal_severity  --warm-start experiments/baseline_binary/best.pt
#   3. multitask         --warm-start experiments/ordinal_severity/best.pt
import os

WARM = {
    'baseline_binary': '',
    'ordinal_severity': 'experiments/baseline_binary/best.pt',
    'multitask': 'experiments/ordinal_severity/best.pt',
}.get(STAGE, '')
```

```
last = f'experiments/{STAGE}/last.pt'

cmd = f'python scripts/train.py --config configs/{STAGE}.yaml'
if os.path.exists(last):
    cmd += f' --resume {last}'
    print('Resuming from', last)
elif WARM and os.path.exists(WARM):
    cmd += f' --warm-start {WARM}'
    print('Warm-starting from', WARM)
print(cmd)
!{cmd}
```

```
02:59:16  INFO     EyeBagModel: encoder=convnext_tiny  params=28.2M  heads=[presence, sever:
02:59:16  INFO     Trainer ready: device=cuda  amp=True  epochs=25  monitor=val_qwk
02:59:41  INFO     Epoch   1/25  train_loss=1.3139  val_qwk=0.0000  lr=2.00e-05  (25s)  ★ b
02:59:46  INFO     Epoch   2/25  train_loss=1.2243  val_qwk=0.0000  lr=3.00e-05  (5s)
02:59:49  INFO     Epoch   3/25  train_loss=1.2058  val_qwk=-0.0593  lr=2.99e-05  (3s)
02:59:51  INFO     Epoch   4/25  train_loss=1.1333  val_qwk=0.0000  lr=2.94e-05  (2s)
02:59:56  INFO     Epoch   5/25  train_loss=1.1728  val_qwk=0.0000  lr=2.88e-05  (6s)
02:59:59  INFO     Epoch   6/25  train_loss=1.1539  val_qwk=0.0000  lr=2.78e-05  (3s)
03:00:02  INFO     Epoch   7/25  train_loss=1.0682  val_qwk=-0.0007  lr=2.66e-05  (3s)
03:00:08  INFO     Epoch   8/25  train_loss=1.0618  val_qwk=0.0096  lr=2.52e-05  (6s)  ★ be
03:00:12  INFO     Epoch   9/25  train_loss=1.0180  val_qwk=0.0837  lr=2.37e-05  (4s)  ★ be
03:00:15  INFO     Epoch  10/25  train_loss=0.9801  val_qwk=0.0623  lr=2.19e-05  (3s)
03:00:23  INFO     Epoch  11/25  train_loss=0.8434  val_qwk=0.0830  lr=2.00e-05  (8s)
03:00:35  INFO     Epoch  12/25  train_loss=0.9596  val_qwk=0.3100  lr=1.81e-05  (12s)  ★ b
03:00:51  INFO     Epoch  13/25  train_loss=0.9010  val_qwk=0.5140  lr=1.60e-05  (16s)  ★ b
03:00:56  INFO     Epoch  14/25  train_loss=0.9069  val_qwk=0.5154  lr=1.40e-05  (5s)  ★ be
03:00:59  INFO     Epoch  15/25  train_loss=0.8839  val_qwk=0.4754  lr=1.19e-05  (3s)
03:01:03  INFO     Epoch  16/25  train_loss=0.8160  val_qwk=0.4951  lr=9.98e-06  (4s)
03:01:09  INFO     Epoch  17/25  train_loss=0.7828  val_qwk=0.5308  lr=8.10e-06  (6s)  ★ be
03:01:12  INFO     Epoch  18/25  train_loss=0.7779  val_qwk=0.4608  lr=6.35e-06  (3s)
03:01:15  INFO     Epoch  19/25  train_loss=0.8386  val_qwk=0.4608  lr=4.76e-06  (3s)
03:01:22  INFO     Epoch  20/25  train_loss=0.7583  val_qwk=0.4622  lr=3.36e-06  (8s)
03:01:25  INFO     Epoch  21/25  train_loss=0.7208  val_qwk=0.4622  lr=2.18e-06  (3s)
03:01:29  INFO     Epoch  22/25  train_loss=0.7758  val_qwk=0.4625  lr=1.24e-06  (4s)
03:01:40  INFO     Epoch  23/25  train_loss=0.7489  val_qwk=0.4798  lr=5.56e-07  (11s)
03:01:42  INFO     Epoch  24/25  train_loss=0.7412  val_qwk=0.4798  lr=1.40e-07  (3s)
03:01:53  INFO     Epoch  25/25  train_loss=0.7915  val_qwk=0.4798  lr=0.00e+00  (11s)
03:01:53  INFO     Early stopping: no val_qwk improvement for 8 epochs. Best was 0.5308 at e
03:01:53  INFO     Training complete. Best val_qwk=0.5308 @ epoch 17
03:01:53  INFO     Checkpoints in experiments/seed_pretrain/  (best.pt, last.pt)
03:01:54  INFO     Final validation metrics:
03:01:54  INFO       accuracy      : 0.9091
03:01:54  INFO       auprc         : 0.9911
03:01:54  INFO       auroc         : 0.9120
03:01:54  INFO       brier         : 0.0648
03:01:54  INFO       dc_accuracy   : 0.6364
03:01:54  INFO       dc_auprc      : 0.8030
03:01:54  INFO       dc_auroc      : 0.7333
03:01:54  INFO       dc_brier      : 0.2246
```

```
03:01:54  INFO       sensitivity    : 1.0000
03:01:54  INFO       specificity    : 0.0000
03:01:54  INFO       within_one     : 0.9455

Next step:
  python scripts/error_analysis.py --checkpoint experiments/seed_pretrain/best.pt --csv data
```

```
# Back up experiments/ to Drive (run after every training cell!)
!mkdir -p {DRIVE_DIR}/experiments && cp -r experiments/* {DRIVE_DIR}/experiments/
!ls -lh {DRIVE_DIR}/experiments/*/
```

```
/content/drive/MyDrive/dermalens/experiments/seed_pretrain/:
total 647M
-rw------- 1 root root 324M Jun 16 03:01 best.pt
-rw------- 1 root root  22K Jun 16 03:01 history.json
-rw------- 1 root root 324M Jun 16 03:02 last.pt

/content/drive/MyDrive/dermalens/experiments/smoke/:
total 263M
-rw------- 1 root root 132M Jun 16 03:02 best.pt
-rw------- 1 root root 1.8K Jun 16 03:02 history.json
-rw------- 1 root root 132M Jun 16 03:02 last.pt
```

```
# Evaluate on the validation split (iterate here; test_internal only at gates).
# seed_pretrain uses data/seed_splits; all real stages use data/splits.
VAL_CSV = 'data/seed_splits/val.csv' if STAGE == 'seed_pretrain' else 'data/splits/val.csv'
!python scripts/evaluate.py --checkpoint experiments/{STAGE}/best.pt --csv {VAL_CSV}
```

```
INFO  EyeBagModel: encoder=convnext_tiny  params=28.2M  heads=[presence, severity, dark_circl
INFO  Loaded experiments/seed_pretrain/best.pt  (epoch 17, encoder=convnext_tiny, severity=Tr
INFO  Dataset loaded: 55 crops from data/seed_splits/val.csv  grade counts: {0: 5, 1: 22, 2:
INFO  Evaluating 55 crops from data/seed_splits/val.csv

=== val.csv | experiments/seed_pretrain/best.pt ===
  accuracy       : 0.9091
  auprc          : 0.9904
  auroc          : 0.9040
  brier          : 0.0671
  dc_accuracy    : 0.5818
  dc_auprc       : 0.8089
  dc_auroc       : 0.7213
  dc_brier       : 0.2304
  dc_ece         : 0.0690
  dc_f1          : 0.7160
  dc_precision   : 0.5686
  dc_sensitivity : 0.9667
  dc_specificity : 0.1200
  ece            : 0.0773
  exact          : 0.6364
  f1             : 0.9524
  mae            : 0.4000
  precision      : 0.9091
  qwk            : 0.5308
  sensitivity    : 1.0000
  specificity    : 0.0000
  within_one     : 0.9636
/content/dermalens/scripts/evaluate.py:122: FutureWarning: Downcasting behavior in `replace`
  if values.replace("", np.nan).dropna().nunique() < 2:

predictions -> experiments/seed_pretrain/predictions_val.csv
metrics     -> experiments/seed_pretrain/metrics_val.json
```

```
Severity confusion (rows=true, cols=pred):
severity_pred  0   1   2  3  4
severity_true
0              0   5   0  0  0
1              0  17   4  1  0
2              0   4  18  2  0
3              0   0   3  0  0
4              0   0   1  0  0
```

```
# Zip the run for local download (error analysis happens on the laptop)
!zip -qr /content/{STAGE}_run.zip experiments/{STAGE}
from google.colab import files
files.download(f'/content/{STAGE}_run.zip')
```

Downloading "seed_pretrain_run.zip":